# Ashim Dahal


Undergraduate Researcher in Computer Vision & Multimodal Learning

codeashim@gmail.com | linkedin.com/in/ashimdahal | Google Scholar | github.com/ashimdahal | ashimdahal.github.io


## Education

**University of Southern Mississippi** Expected May 2027
*B.S. in Computer Science; GPA: 3.87/4.0; Keystone Honors Scholar* *Hattiesburg, MS*

## Selected Publications

**[C1]** **A. Dahal**, A. Ghimire, S. A. Murad, and N. Rahimi, "POVQA: Preference-Optimized Video Question Answering with Rationales for Data Efficiency," ***CVPRW***, Proceedings Track, 2026.

**[C2]** **A. Dahal**, S. A. Murad, and N. Rahimi, "Embedding Shift Dissection on CLIP: Effects of Augmentations on VLMs' Representation Learning," ***CVPRW***, Proceedings Track, 2025.

**[J1]** **A. Dahal**, S. A. Murad, and N. Rahimi, "Heuristic Comparison of Vision Transformers Against Convolutional Neural Networks for Semantic Segmentation on Remote Sensing Imagery," ***IEEE Sensors Journal***, 2025.

## Research Experience

**MCTS-VQA: Monte Carlo Tree Search for Video Evidence Collection** Jan 2026 – Present
*Undergraduate Researcher*
***Advisor:*** *Bikramjit Banerjee*

- Developing a Monte Carlo Tree Search framework for identifying question-relevant frame evidence in long-form video question answering.

**Adaptive Anchor Policies for Efficient 4D Gaussian Streaming** Apr 2025 – Jan 2026
*Undergraduate Principal Investigator*
***Advisors:*** *Rabab Abdelfattah and Nick Rahimi*

- Developed adaptive anchor-management policies for efficient streaming of dynamic 4D Gaussian representations while preserving temporal and rendering quality.
- Independently proposed and led the project through a competitive **$5,500 DCUR Summer Research Grant**.

**Cyber Innovations Lab, University of Southern Mississippi** Aug 2023 – Present
*Undergraduate Research Assistant (fully funded); Advisor: Nick Rahimi* *Hattiesburg, MS*

- Led development of **POVQA**, a rationale-guided preference-optimization framework that reduced visual input tokens by 40% while retaining more than 95% of baseline answer accuracy [**C1**].
- Conducted representation-learning and remote-sensing studies involving CLIP, Vision Transformers, and convolutional networks, resulting in [**C1,J1**].
- Co-designed research methodology for a **$51,000 NASA EPSCoR project** on efficient vision-language models

## Selected Honors & Leadership

**$1,500 Eagle SPUR Research Grant** – 3DGS and Robotics Research 2026
**$500 Best Presentation on Computational Approaches** – Undergraduate Research Symposium 2026
**$5,500 DCUR Summer Research Grant** – Student Principal Investigator 2025
**Lead Organizer** – Google Developer Group On Campus at USM 2025 – Present

## Technical Skills

**Languages** : Python, C++, C#, SQL, JavaScript, CUDA, Bash
**ML/DL** : PyTorch (DDP, Lightning, FSDP), TensorFlow, JAX, Hugging Face Transformers, scikit-learn
**Computer Vision** : Vision Transformers, CLIP, 3D Gaussian Splatting, NeRFs, Video QA, Image Segmentation, Multiview Stereo, Object Detection
**Infrastructure** : CUDA, Multi-GPU Training, SLURM, HPC, Docker, FastAPI, Git, Weights & Biases, Linux